

Lab 1 - Functional Tests for the Demonstration

Requisitos de conclusão

The demonstration of the Lab 1 work includes running a series of functional tests with the protocol. The tests are performed in real serial ports of the laboratory. The functional tests and their procedure are listed below:

Test 1: File Transfer

1. Start the receiver.
2. Start the transmitter.
3. Wait until the transmission finishes.
4. Check the integrity of the file received.

Test 2: File Transfer after One Timeout

1. Start the transmitter.
2. Wait a few seconds, in order to trigger one timeout in the transmitter.
3. Start the receiver.
4. Wait until the transmission finishes.
5. Check the integrity of the file received.

Test 3: File Transfer without a Receiver

1. Start the transmitter.
2. Do NOT start the receiver.
3. Check what happens.

Test 4: File Transfer with Retransmissions

1. Start the receiver.
2. Start the transmitter.
3. During the transmission, disable the channel for a few seconds.
4. Reenable the channel after a few seconds.
5. Check the integrity of the file received.

Test 5: File Transfer with Random Noise

1. Start the receiver.
2. Start the transmitter.

3. During the transmission, add random noise to the channel.
4. Disable the random noise after a few seconds.
5. Check the integrity of the file received.
6. Check whether REJ frames have been sent.

Test 6: File Transfer with Duplicate Frames

1. Start the receiver.
2. Start the transmitter.
3. During the transmission, disable the Rx -> Tx channel, in order to disable the reception of the RR frames.
4. Reenable the Rx -> Tx channel after a few seconds.
5. Check the integrity of the file received.